CS3 RESEARCH EXPERIENCE FOR TEACHERS 2024

Training, Validation, and Testing in Machine Learning

Why we split our data, and what each part is for

Mehmet Kerem Turkcan

Center for Smart Streetscapes, Columbia University

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OVERVIEW

One dataset, three jobs

Training data

Teaches the model patterns and relationships.

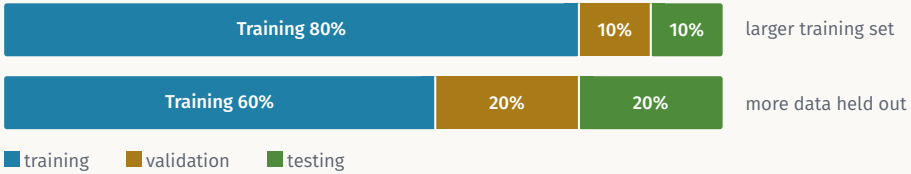
Validation data

Tunes hyperparameters and helps choose between models.

Testing data

Evaluates the final model, once, on data it has never seen.

The two ends of the typical range given on the next slides, drawn to scale:



Why split your data?

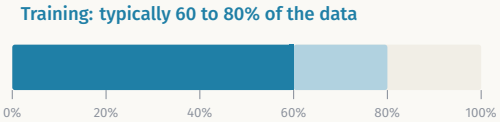
- To evaluate model performance properly.
- To avoid overfitting.
- To make sure the model generalizes to new, unseen data.
- To make informed decisions about model selection and hyperparameters.

The idea in one line

A model should be judged on data it did not learn from. Splitting the dataset sets that data aside before training begins.

Training data teaches the model

- The largest portion of your dataset.
- Used to teach the model patterns and relationships.
- The model learns by minimizing its loss on this data.



Dark segment: the low end of the range. Light segment: the rest of the typical range.

Watch out

Training data alone can lead to overfitting: a model can score well on examples it has already seen and still fail on new ones.

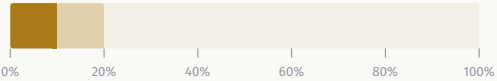
Validation data tunes the model

- A smaller portion of your dataset.
- Used to tune hyperparameters and to select between models.
- Gives an unbiased evaluation during development, because the model does not train on it.
- Helps prevent overfitting to the training data.

Hyperparameter

A setting chosen before training rather than learned from the data, such as the learning rate or the number of epochs.

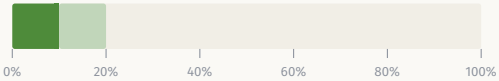
Validation: typically 10 to 20% of the data



Testing data evaluates the final model

- The final portion of your dataset.
- Used only once, to evaluate the final model.
- Simulates real-world performance on unseen data.
- Crucial for estimating how well the model truly generalizes.

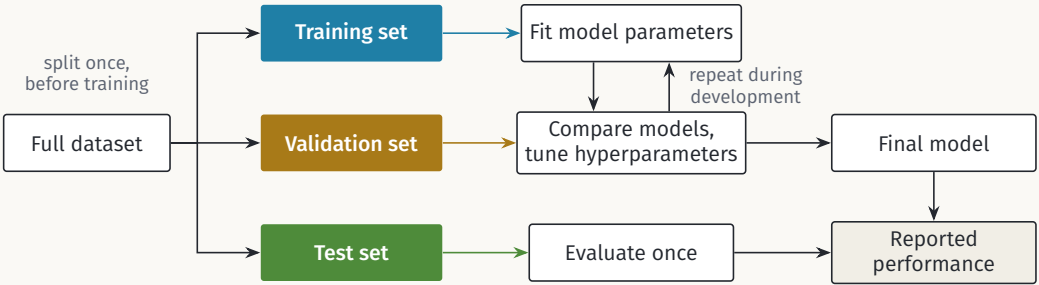
Testing: typically 10 to 20% of the data



Classroom analogy

Training data is the homework, validation data is the practice quiz, and testing data is the final exam: students see it once, and only after they have finished studying.

Where each split is used



Training teaches, validation tunes, testing evaluates. Training and validation data are used again and again during development. The test set stays sealed until the end and is used once.

IMPORTANCE

Why this split is important

Prevents overfitting

The model is rewarded for generalizing beyond the training data, not for memorizing it.

Simulates real-world scenarios

The test set plays the role of truly unseen data.

Enables fair model comparison

Every model is scored on the same test set.

Builds trust in model performance

The final number comes from an unbiased evaluation.

IMPORTANCE

Common pitfalls to avoid

Data leakage between sets

Information from the validation or test data finds its way into training, so the evaluation is no longer on unseen data.

Using test data more than once

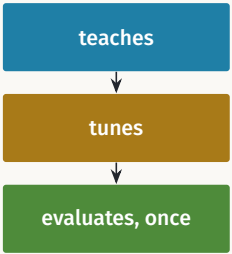
Each look at the test score that influences a decision makes it a little less of a final exam.

Overfitting to the validation set

Tuning against the same validation set again and again can fit its quirks as well.

Ignoring dataset shift or time

The data a model meets after deployment may differ from the data it was split from, for example when it is collected later.

Key takeaways

- Proper data splitting is crucial for model development.
- **Training** data teaches, **validation** data tunes, and **testing** data evaluates.
- This approach leads to more robust and reliable models.
- Always keep the test data separate, and use it only once.